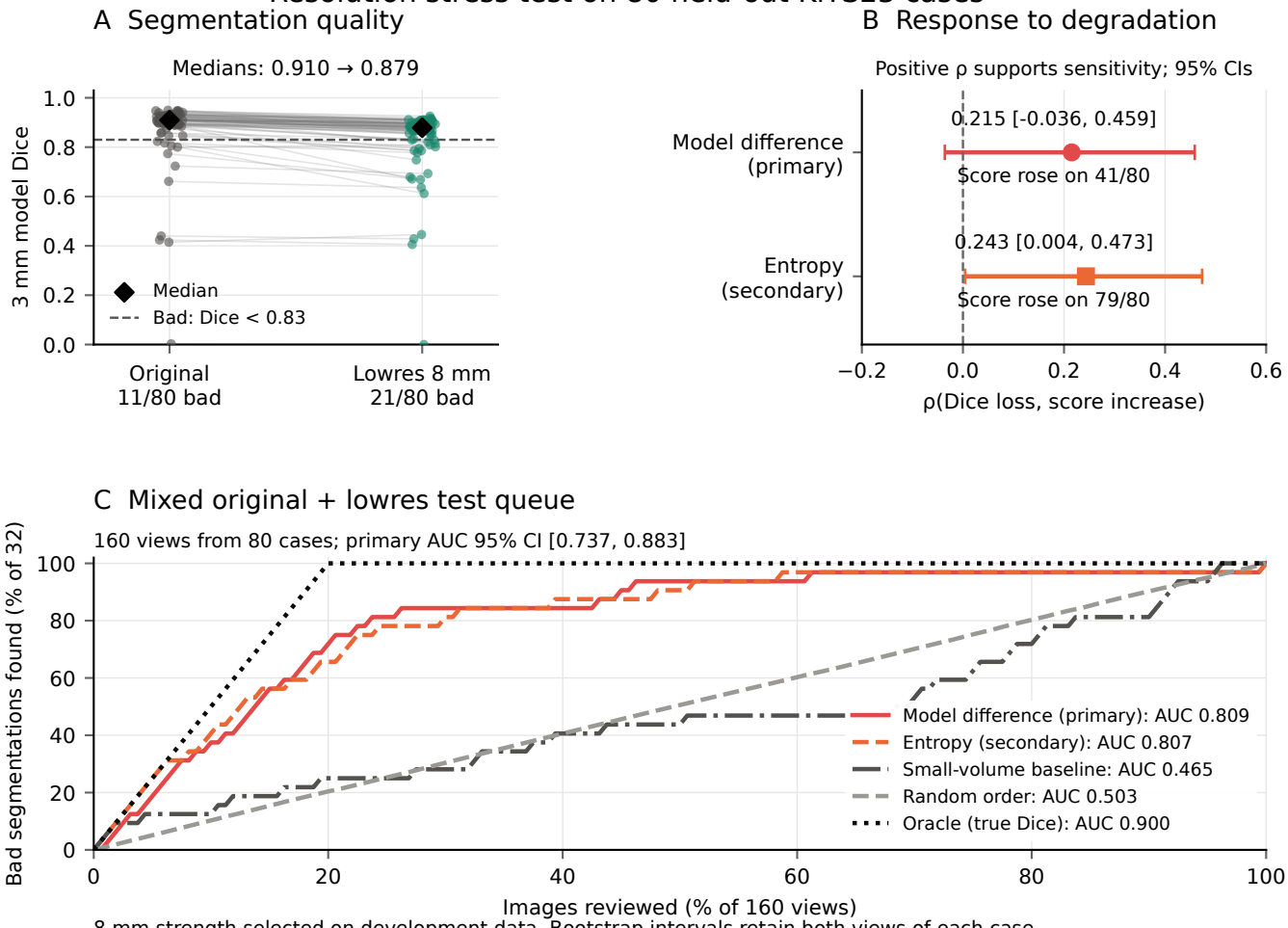
Resolution stress test on 80 held-out KiTS23 cases
A Segmentation quality
Medians: 0.910 → 0.879
3 mm model Dice
1.0
0.8
0.6
0.4
0.2
0.0
Median
Bad: Dice < 0.83
Original
11/80 bad
Lowres 8 mm
21/80 bad
B Response to degradation
Positive ρ supports sensitivity; 95% CIs
Model difference
(primary)
0.215 [-0.036, 0.459]
Score rose on 41/80
Entropy
(secondary)
0.243 [0.004, 0.473]
Score rose on 79/80
−0.2
0.0
0.2
0.4
0.6
ρ(Dice loss, score increase)
C Mixed original + lowres test queue
160 views from 80 cases; primary AUC 95% CI [0.737, 0.883]
Bad segmentations found (% of 32)
100
80
60
40
20
0
0
20
40
60
80
100
Images reviewed (% of 160 views)
Model difference (primary): AUC 0.809
Entropy (secondary): AUC 0.807
Small-volume baseline: AUC 0.465
Random order: AUC 0.503
Oracle (true Dice): AUC 0.900
8 mm strength selected on development data. Bootstrap intervals retain both views of each case.